



C & I MAINT Department
SIMHADRI

Ref: SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Date:

18/06/2022(16:23:07)

Subject: REVISION OF LMI ON HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING REQUIREMENTS

Revision of [OGN-OPS-C&I-003](#) was approved on 02.06.2022 to include the following points:

- OGN Heading changed from "HANDLING OF ELECTROSTATIC SENSITIVE DEVICES" to "HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING REQUIREMENTS"
- Records to be maintained- Table contents reviewed, and frequency column added.
- Annexures- Draft Format of Records to be maintained added

To include the above changes LMI on 'HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING REQUIREMENTS' is being revised.

Approved [OGN-OPS-C&I-003](#) and draft LMI is attached for reference.

Put up to competent authority for kind approval of the revised LMI.

Submitted by:

अजीतकुमार रौशन - वरिष्ठ प्रबंधक

AJEET KUMAR RAUSHAN - 103000 (SENIOR MANAGER ,C & I MAINT) ,Simhadri
(AKRAUSHAN@NTPC.CO.IN, 9434077809)

On: 18/06/2022(16:23:07) **Ref:** SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:18/06/2022(17:08:15) **Ref:**SMTPP/C & I MAINT/2022-23/C&I/LMI/195567

- May please approve.

Submitted by:

चौ. किशोर कुमार - अपर महाप्रबंधक

Ch Kishore Kumar - 006106 (ADDL. GENERAL MANAGER ,C & I MAINT)
(CHKISHOREKUMAR@NTPC.CO.IN, 9440918196) ,Simhadri

On:20/06/2022(11:26:09) **Ref:**SMTPP/C & I MAINT/2022-23/C&I/LMI/195567

1. Para 9.0 Documents related to Sl Nos 1,2,3 and 4 may be collected and annexed
2. Details of VMS panels, various PLCs in offsite areas (including IMWB) may be mentioned in the Anneure-3 for which yearly/ during overhauling Resistivity measurements have to be done and records to be maintained

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:20/06/2022(18:21:26) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Submitted by:

चौ. किशोर कुमार - अपर महाप्रबंधक

Ch Kishore Kumar - 006106 (ADDL. GENERAL MANAGER ,C & I MAINT)
(CHKISHOREKUMAR@NTPC.CO.IN, 9440918196) ,Simhadri

On:24/06/2022(16:13:13) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

AS BUILD DRAWING OF EARTHING AND GROUNDING SCHEME & LAYOUT FOR MAIN PLANT AND OFFSITE AREA HAVE BEEN UPLOADED IN C&I SITE UNDER KNOWLEDGE MANAGEMENT TAB.

FOLLOWING LINK CAN BE USED TO SEE THE DETAILS:

<http://10.0.176.1:7001/ci/folder.php?id=1143>

Submitted by:

अजीतकुमार रौशन - वरिष्ठ प्रबंधक

AJEET KUMAR RAUSHAN - 103000 (SENIOR MANAGER ,C & I MAINT)
(AKRAUSHAN@NTPC.CO.IN, 9434077809) ,Simhadri

On:29/06/2022(10:29:45) **Ref:**SMTPP/C & I MAINT/2022-23/C&I/LMI/195567

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:29/06/2022(12:58:12) **Ref:**SMTPP/C & I MAINT/2022-23/C&I/LMI/195567

Submitted by:

चौ. किशोर कुमार - अपर महाप्रबंधक

Ch Kishore Kumar - 006106 (ADDL. GENERAL MANAGER ,C & I MAINT)
(CHKISHOREKUMAR@NTPC.CO.IN, 9440918196) ,Simhadri

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LMI Reviewed and forwarded for further review and approval

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:01/07/2022(17:12:08) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Submitted for approval of competent authority

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
(JHIMLIDHAR@NTPC.CO.IN, 9650998085) ,Simhadri

On:02/07/2022(10:35:01) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Approval may please be accorded.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:16/07/2022(19:32:04) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
(RKJHA02@NTPC.CO.IN, 9415342046) ,Simhadri

On:17/07/2022(10:37:39) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri

On:18/07/2022(12:11:53) **Ref:**SMTTP/C & I MAINT/2022-23/C&I/LMI/195567

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LOCATION MANAGEMENT INSTRUCTION

No. LMI/OGN/GEN/C&I/003

ISSUE No.: 1

REV: 07

DATE: June 2022

HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING REQUIREMENTS

**Ref: OPERATION GUIDANCE NOTE: COS-ISO-00-OGN/OPS/C&I/003 REV-03 on
"Handling of Electrostatic Sensitive Devices & Grounding Requirements"**

NTPC Ltd. Simhadri, Visakhapatnam

(For Internal Circulation Only)

SMPP-ISO-00-LMI/OGN/OPS/C&I/003

Rev No: 7, June 2022

NTPC Limited

Approved for

Implementation by:

HOP (Simhadri)

Date:

SMPP, NTPC Limited**NATIONAL THERMAL POWER CORPORATION****SIMHADRI****OPERATION & MAINTENANCE DEPARTMENT**

LMI (Name):	HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING REQUIREMENTS		
LMI No.:	LMI/OGN/GEN/C&I/003		
Issue No.	1	Revision: 07	Date: June2022

	Name	Designation	Signature
Prepared by	Ajeet K Raushan	Sr Manager(C&I)	
Checked by	Jayamol Rachel V	AGM (C&I)	
Reviewed by	Ch Kishore Kumar	AGM(C&I)	
Reviewed by	T.R.Rajagopalan	HOD (C&I)	
Reviewed by	Mathew E Kovoov	GM(M)	
Reviewed by	R K Jha	GM(O&M)	
Approved by	G C Choukse	HOP (Simhadri)	

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**HANDLING OF ELECTROSTATIC SENSITIVE DEVICES &
GROUNDING REQUIREMENTS**

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SMPP, NTPC Limited**HANDLING OF ELECTROSTATIC SENSITIVE DEVICES & GROUNDING
REQUIREMENTS****1.0 INTRODUCTION:**

This document contains the LMI (Location Management Instruction) which fulfill the requirement of directives as per OD: GEN: 001 on the subject.

- a) Modern micro-electrostatic circuits found in DDC Systems, Maxdna Systems, PLCs, computers, telecommunication & protection equipment can be damaged by static electricity unless suitable precautions are taken.
- b) Properly designed circuits will normally be safe from damage when housed in the equipment case supplied by the manufacturer, but may be subject to damage when handled.

This LMI note gives practical advice for personnel who may need to handle such circuits during Operation & Maintenance of such circuits/devices.

2.0 SCOPE

The LMI describes the precaution to be taken when handling and testing all kinds of electrostatic sensitive components, circuits and assemblies.

3.0 RECORDS OF REVISION

SUPERSEDED DOCUMENTS: LMI/OGN/GEN/C&I/003, REV: 06

Details for this revision are as below :

- a. OGN Heading changed from "Handling of Electrostatic Sensitive Devices" to "Handling of Electrostatic Sensitive Devices & Grounding Requirements"
- b. Records to be maintained- Table contents reviewed, and frequency column

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added.

c. Annexures- Draft Format of Records to be maintained added

4.0 MECHANISM OF FAILURE

- a) Most people have experienced a small electric shock, caused by static electricity, when touching a metallic object, such as filing cabinet, after walking across a carpet. The potential generated in human body or in any object (if not properly grounded) may be several thousand volts and had it been discharged into an unprotected electronics circuits, components within that circuit could be seriously damaged. The damage will however not always be immediately apparent but may cause failure at a later date.
- b) Semiconductor junctions within the integrated circuits are very small and as the usual working voltage will be low, the material layers from which they are made will be correspondingly very thin. In the future, as junction thickness decreases further to permit higher circuit packing density and faster operating speed, it is likely that devices may become more susceptible to damage by static charge.

5.0 MARKING OF ELECTROSTATIC SENSITIVE DEVICES

- a) IS 10087-1981 Code of Practice for Handling of Electrostatic Sensitive Devices", defines a label, identical to that in IEC Publication 747-1, to indicate electrostatic sensitive devices.
- b) The symbol is a yellow hand on a black triangular background and will be found both on packing material and on the case of the device itself. It may be necessary to remove the front cover of a piece of equipment, when in service, to see the symbol.
- c) Manufacturer's literature supplied with the device should be also indicate that the device is sensitive to static charge.

SMPP, NTPC Limited**6.0 SPECIAL HANDLING PROCEDURE FOR SITE WORK:**

If it becomes necessary to withdraw module containing an electrostatic sensitive device from the equipment case, the following precautions will reduce the possibility of damage.

- a) Before removing a module, ensure that you are at the same electrostatic potential as the module by touching the equipment case. Care should be taken if the equipment is energised and the use of a wrist strap as outline in (f) below is recommended.
- b) Handle the module by its front plate, frame, or edges of the printed circuit board. Avoid touching the electronic components, printed circuit track and connectors.
- c) Do not pass the module to any person without first ensuring that you are both at the same electrostatic potential. Shaking hands achieves equipotential.
- d) Place the module on an antistatic surface, or on a conducting surface which is the same potential as yourself.
- e) Store or transport the module in a conductive bag. Conductive bags are widely available but in case of difficulties they can be obtained from the equipment manufacturer.
- f) If you are making measurements on the internal electronic circuitry of (for example) on amplifier in service, it is recommended that you are earthed to the case with a conductive wrist strap. Wrist straps should have a resistance to ground between 500 k ohms to 10 M ohms. If a wrist strap is not available, you should maintain regular contact with the case to prevent the built up of static electricity. Instrumentation which may be used for making measurements should be earthed to the case whenever possible.

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- g) Voltage for continuity testing greater than 1.5 volts should be used with caution. Any tests should be made with a low voltage open circuit transient free tester with a short circuit current of less than 10 mA. Care should be taken with the use of multi range test meters. Simple audible (buzzer or bell) types of continuity tester should not be used when electrostatic sensitive devices are present.

7.0 SPECIAL HANDLING PROCEDURE FOR STORES / LABORATORY / WORKSHOP

It is recommended that a Special Handling Area should be established as defined in - IS :10087 - 1981. (Reaffirmed Year: 2015).

8.0 GROUNDING:

Grounding of panels & electronic earthing is preliminary required for basic safety of human and control equipment. A strict adherence is required to prevent shock hazard, protection against lightening, Electrostatic discharge (ESD) hazards, Electromagnetic Interference (EMI).

- a) Isolation to be ensured for electronic shield grounding and panel grounding.
- b) Ensure grounding scheme of the C&I system is followed as per the approved drawing/design.
- c) Electronic Earth pit/Grid resistance to be measured in every Unit O/H by 3-point (fall-of-potential) method. Value <1 Ohm.
- d) Proper tightness of terminations at earth pit/grid raiser end terminations covered with grease or bitumen paint (to avoid rusting due to harsh environmental conditions).
- e) Panel & Electronic ground resistance of each panel to be maintained (less than <1 Ohm).

SMPP, NTPC Limited**9.0 RECORDS TO BE MAINTAINED:**

Sl. No.	Record Description	Maintained by	Frequency
1	As build drawing of Panel Earthing of each DDCMIS / PLC system / UPS / Charger / any other Panel with Electronic cards/ Antistatic work bench	Head of C&I	One time
2	As build drawing of 24V DC negative Earthing of each Charger DCDB	Head of C&I	One time
3	As build drawing of Neutral Earthing of each UPS ACDB	Head of C&I	One time
4	As build drawing of Electronic Earthing of each DDCMIS / PLC system / Charger DCDB / VMS / any other Panel with Electronic cards/ Electronic Earthing JB / Instrument Ground Bus (IGB) Test Assembly/	Head of C&I	One time
5	Record of Resistivity - Earth Connection of Anti-Static Work benches (Annexure-1)	Head of C&I	Overhaul
6	Record of Resistivity – Antistatic Wrist Bands of CER panels (Annexure-2)	Head of C&I	Overhaul
7	Earthing checking during 1st charging of the DCS panels / VMS Panels/ UPS / Chargers	Head of C&I	During R&M / New installations
8	Record of Earth pit/ Earth Grid resistance in unit O/H before and after Earth pit /Grid maintenance (Annexure -3)	Head of C&I	Overhaul
9	Record of panel & Electronic resistance of each panel in CER's after maintenance work in Unit O/H (Annexure-4)	Head of C&I	Overhaul

10.0 REFERENCE DOCUMENTS

- a) IS:10087-1981(Reaffirmed Year: 2015): Indian standard Code of Practice for Handling of Electrostatic Sensitive Devices".
- b) IEC Publication 747-1:1983: Semiconductor Devices: Discrete Devices and Integrated Circuits, Part- 1: General.
- c) IS 3043 - 2018: Code for Earthing Practice
- d) 1100-2005 IEEE Recommended Practice for Powering and Grounding Electronic Equipment
- e) 1050-2004 IEEE Guide for Instrumentation and Control Equipment Grounding in Generating Stations

SMPP, NTPC Limited**11.0 ANNEXURES (Draft Format of Records to be Maintained)**

Annexure -1

Antistatic Work Bench Number	Resistivity (Ω)	Remarks

Annexure -2

Antistatic Wrist Bands Location	Resistivity (Ω)	Remarks

Annexure -3

Earth Raiser / Earth Pit Number	Resistance before Maintenance (Ω)	Resistance After Maintenance(Ω)	Remarks

Annexure -4

Panel	Panel Earth Resistance (Ω)	Electronic Earth Resistance (Ω)		Electronic Shield to Panel Shield (mV)	Remarks
		Cable-1	Cable-2		